

Final plan: Business Entity Resolution (Team Grenuke)

Status: accepted, v1.0, Fri 25 Sep 2026, 14:15 IST. The coordinator (Ameya) chose this plan from Plan A (plans/ameya/PLAN.md) and Plan B (plans/sachi/PLAN.md). Plan C was not submitted. The decision record, grafts and rejected ideas are in plans/DECISION.md. **Changes after v1.0:** each change gets a record in docs/decisions/. Only the coordinator edits this file. The version history is at the end.

- **Deadline:** Sun 27 Sep 2026, 23:59 IST.
- **Leaderboard budget:** 5 uploads per day (15 in total).
- **Metric:** macro F0.5 per Source-1 (S1) entity, singletons included.
- **Tags used below:**
 - **[M]:** measured on the provided data;
 - **[F]:** a fact from the problem statement;
 - **[H]:** a hypothesis. Every hypothesis has a gate in §9 that can overturn it.

0. TL;DR

- **Pipeline.** Eight stages, each cached and vectorized:
 - records → normalize → block (multi-view retrieval + exact keys) → pair features;
 - stage-1 GBDT → stage-2 collective model with calibration;
 - ownership (who owns each S2/S3 record) → per-entity F0.5 decision → outputs.
- **Why this shape [M]:**
 - S1 names collide constantly: 46–54% share an exact name with another S1. Name and address together almost never collide.
 - Every S2/S3 record has at most one owner.
 - A false merge costs about 3 times a miss.
 - The main false-positive trap is the **look-alike**: the same street, a nudged house number, and a business-changing word ("Exports", "Holdings").
 - Test has **23% more S2/S3 records per S1** than train.
- **How we work.**
 - Friday is a **v0 baseline**. Each later step is an increment that has to pass its gate.
 - Every complex component must beat a simpler one on the shared holdout, using a paired bootstrap (§9). This rule comes from Plan B.
 - Every claim carries a tag.
- **First valid submission:** Fri 25 Sep before 23:30 IST.

1. Evidence ledger

#	Fact	Tag	Consequence
1	Train: S1 2,206,821, S2 5,034,616, S3 5,285,603 (US 60%, India 40%). Test: S1 1,732,544, S2 4,887,273, S3 5,082,316 (India 47%, US 38%, France 15%). Test labels are exactly {US, India, France}.	[M]	Everything is vectorized or compiled, processed per country. <code>country</code> is an open set.
2	Each S2/S3 record matches at most one S1: 0 violations in 7,638,365 true pairs.	[M]	Competition between S1s for a record is a strong precision signal (ownership, §4.7).
3	No cross-country matches.	[M]	Block within a country label only.
4	Matches per S1: mean 3.46, max 11. Distribution: 0: 5.6%, 1: 5.4%, 2: 17%, 3: 24%, 4: 22%, 5: 14.6%, 6: 7.5%, 7+: 4%.	[M]	The job is mostly "find all 3–4". Getting the set size right matters more than singletons.

#	Fact	Tag	Consequence
5	26% of S2/S3 records match nothing (S2 26.6%, S3 25.3%). We call them orphans.	[M]	There are many hard negatives, and they have to be modelled.
6	Exact core-name collisions: 46% of US S1 and 54% of India S1 share a name with another S1. Exact normalized address collisions: about 5% (largest group 14). Both at once: 6 S1.	[M]	Never trust the name alone. Name-rarity and shared-address counts become features.
7	True pairs: weak name 15.9% (India 26.1%), weak address 4.7%, both weak 0.09%. The S2/S3 address is empty in 4.4% of pairs. Weak address with more than 3 tokens: only 0.27%.	[M]	Address-driven plus name+address retrieval recovers almost everything. Name-only retrieval is needed only for records with empty or short addresses.
8	House numbers in true pairs: first number equal 83% (US 88%, India 77%), number sets equal 73%, some overlap 95%.	[M]	Use number <i>relations</i> (truncation, suffix, leading zeros, nudges), not exact equality.
9	Look-alike signature. Orphans close to an S1 on name and street words: the first number is equal in only 12% (true pairs 85%), and 77% add a name word (true pairs 23%). The added words are business-changing: <i>group, holdings, industries, public, enterprises, exports, infratech, ventures, overseas</i> . In true pairs they are noise: <i>center, services, dba, shri/sri/smt, dr/mr, formerly, incorporated</i> .	[M]	Number-relation features and extra-token features (with learned per-token distractor rates) are core, not optional.
10	Postcodes are essentially absent: $\leq 0.5\%$ of addresses in US, India and France .	[M]	No postcode field or feature.
11	Indic-script S2/S3 names appear in 18% of India true pairs (7% overall). Accents are injected in every country.	[M]	Transliteration plus name variants (§4.2).
12	Noise catalogue (Plan A §1.3): typos and OCR swaps, word shuffles, legal-form changes, domains and hashtags, initials, DBA forms, invented brand names at the true address, appended phones, reordered address parts, abbreviations, state as code/name/script, neighbouring cities, <code><NULL></code> , <code># / N° /bis/ter</code> , wrong ordinals, India numbering (H.No/Plot/Door No), truncated numbers, fragments.	[M]	These drive the normalizer and features.
13	Test shift: 5.75 S2/S3 per S1 in test vs 4.68 in train (India 5.82, US 5.76, France 5.53).	[M]	Thresholds tuned on train may not transfer. We need test diagnostics (§4.7, gate G8).
14	The extra test records are the same kinds as train's , not records whose S1 owner is missing. In India, where train and test S1 pools are similar in size, how close records sit to their nearest S1 is unchanged: close on name and address 31.7% vs 31.8%; nothing close 43% vs 44%. Removing owners would give about 26% and 53%.	[M] (crude proxy)	So test has more matches and/or more look-alikes per S1. A "drop S1s" stress test models the wrong shift. It checks one failure mode only.
15	France follows the same generator as US and India: the same closeness profile (30.6% close on both), <code>R / AVE / IMP .</code> , <code>N°16</code> , <code>bis/ter</code> , region $\leftrightarrow$ department swaps, generic names (Amis, Club, Union...).	[M] (qualitative)	Use country-agnostic features plus French lexicons (§6).
16	No ID-order leakage (correlation -0.001). A few orphans look like true matches (label noise).	[M]	Never use IDs or row order. Expect a small irreducible error.

The measurements for #6–#10, #13 and #14 were made on 25 Sep; method and numbers are in `docs/handover/2026-09-25_1414_ameya_final-plan.md`. The rest comes from Plan A §1 and Plan B §1.

2. What the metric implies

Per entity, $F = 1.25 \cdot c / (0.25 \cdot |T| + |P|)$, where c is the number of correct predictions, T the true set and P the predicted set. An empty prediction scores 1 on a singleton and 0 otherwise.

Break-even calibrated probabilities, assuming independence:

situation	include the candidate if p >
only candidate (singleton vs 1 match)	0.50
2nd candidate, the 1st is certain	0.73
4th candidate, 3 are certain	0.77

- **A false merge costs about 3 times a miss**, and the bar rises with set size. Calibration and high-confidence precision decide the score.
- **Recall still matters**, because the mean |T| is 3.46.
- **Decision rule candidates** (compared at gate G6):
 - a tuned global threshold (simple);
 - an exact expected-F0.5 per S1. Candidates are sorted by p. The expected F is computed for every prefix length with a Poisson-binomial dynamic program over "true inside" and "true outside". It costs almost nothing in numba.

3. Pipeline

stage	module	writes (docs/CONTRACTS.md)	contract
00 records	ber.records	work/records/{train,test}.parquet, work/records/truth.parquet	C3
01 normalize	ber.normalize	work/norm/<tag>/<split>.parquet	C3
02 block	ber.block	work/candidates/<tag>/<split>.parquet + blocking report	C4
03 features	ber.features	work/features/<tag>/<split>.parquet	C8
04 train	ber.model	models + OOF stage-1 scores work/scores/<tag>-s1/train.parquet	C5
05 predict	ber.model	work/scores/<tag>/<split>.parquet (calibrated p)	C5
06 decide	ber.model	work/matches/<tag>/<split>.parquet	C9
07 write	ber.outputs	output/matching_results.tsv, output/candidate_pairs.tsv	C6
08 evaluate	ber.eval	work/reports/<tag>.json (holdout, per country)	C7

- One CLI: `python -m ber.pipeline --stage <stage|all> --split <train|test> --tag <tag> [--in <stage>=<tag>]`.
- Each stage reads the previous stage's artifact by tag, so any stage can be re-run, swapped or A/B-tested alone.
- Strings exist only up to stage 01. After that, records are int64 `eid`s and int32 token codes.

4. Stage design

4.1 Records (00): v0 is in the skeleton PR

`ber.io` readers go in, and out come `eid`, `source`, `country`, `name` and `address` per record, plus the truth pairs (train). The cache takes about 1 minute to build.

4.2 Normalization (01)

v0 (Friday)

- **Unicode:**
 - NFKC, then Indic→Latin, then NFKD with combining marks dropped, then lowercase. This removes real and injected accents.
 - Indic→Latin uses our own offset table (Plan A). The 9 Indic blocks share the ISCII-derived layout, so one table handles virama, vowel signs, anusvara and nukta, with schwa deletion.
 - `&/+` become "and". Junk symbols are stripped. Bracket contents are kept.
- **Names:**
 - `n_core`: tokens without legal forms, honorifics or junk.
 - `n_legal`: a canonical code (llc, inc, corp, ltd, pvt_ltd, llp, lp, pllc, pc, sa, sas, sasu, sarl, eurl, sci, snc, ei, ...).
 - `n_concat`: the core with spaces removed.
 - Flags: domain, hashtag, phone, DBA (both parts kept), Indic, honorific, token count.
- **Addresses:**

- Remove null tokens.
- Canonicalize numbering markers: h.no, door no, plot, flat, site, sf, sy, no., n°, #, unit, apt, suite, pmb, po box.
- Ordinals: 7nd→7, tenth→10.
- Street types get one short form across US, India and France: st, ave/av, rd, dr, ln, blvd/bd, ct, pl, trl, marg, nagar, colony, sector, rue/r, ch, imp, allee, quai, rte, fg. "saint" becomes st on both sides.
- States: US names ↔ codes, India states ↔ codes, **France department** → **region** (hand-written, documented).
- Numbers: every digit run, with leading zeros stripped and letter suffixes split off ("8444b" → 8444 + b; bis/ter treated as suffixes). Composite ids are kept whole and in parts. Separate fields for the primary number and the unit.
- `a_street` (street tokens) and `a_city` (the component before the state).
- Flags: empty, **short (≤3 tokens)**, landmark (near/opp/behind/nr/près/face), PO box, null, fragment.
- France: drop "Cedex"; keep arrondissements ("8e") as a number with a flag.
- **No postcode field** (#10).
- **Performance:**
 - Regex over 24 processes: about 5 minutes for 23.7M records.
 - Tokens are encoded once into int32 vocabularies with IDF, per country, fitted on train and test together. This is unsupervised and adapts to France.

v1 (Saturday, gate G12)

- Learned Indic token dictionary from train pairs (Plan A). Each Indic token maps to the most similar Latin token in the paired S1 name. Accepted with ≥2 supporting pairs and ≥60% share, and **fitted without the holdout**.
- **Consonant skeleton** of names (Plan B), for proper nouns the dictionary can't cover.
- **OCR variant** of names (Plan B): 5→s, 0→o, 1→l inside words. It sits next to the original, and features take the maximum.
- Domain stem uses only tokens of 3+ characters (Plan B guard). Learned city aliases are fitted without the holdout.

4.3 Blocking (02): sets the recall ceiling

- **Partition** by the exact country label. A record with an unseen or empty label is searched against all partitions. That is defensive only: this test set has just three labels.
- **Views**. Each is a TF-IDF fitted per country on train+test.

view	text	why
V-both	name core + street/number/city, char 3-grams	the main retriever; the address breaks ties between common names (#6)
V-addr	address char 3-grams + number/street word tokens	weak names: scripts, domains, invented brands (#7)
V-name-short	name char 2–4-grams, only over S2/S3 records with empty or ≤3-token addresses (about 5% of records)	weak or empty addresses (#7). Replaces Plan A's name view over all records (Plan B's design)
keys (exact, bucket-capped)	(primary number, first street word); domain stem; DBA inner name	cheap and targeted

- **Search engine (gate G2)**. TF-IDF, then SVD to 256 dimensions (fitted on a 2M sample), then L2-normalized. Then an exact GPU top-k over fp16 tiles, with running top-k buffers along **both** axes: S1→R top-K and R→S1 top-k in one pass (Plan A).
 - The CPU fallback is sparse exact top-k with the highest-frequency n-grams pruned.
 - Measure recall@K of SVD vs sparse on a 100k-entity sample before scaling up.
- **Trim**. The union carries metadata per view: score, rank in the S1 list, rank in the R list, and a view mask.
 - **v0 uses a heuristic trim**: keep the top N per S1 by best view score (N≈30), plus the top 2 per R.
 - A learned pre-ranker (Plan A) comes in only through gate G11: when pairs per S1 or feature time demand it, and only if it loses less than 0.1% of the union's recall.

- **K** is fixed in v0 (S1→R 40 per view, R→S1 8), then tuned on the recall-vs-pairs curve. A view or key stays only if it adds **at least 0.1 pt** of holdout pair recall (Plan B).
- **Acceptance gate G1:**
 - holdout pair recall **≥99.0% (v0)** and **≥99.5% (v1)**, overall and per country and source;
 - a recall report for hard cases (Indic, domain/hashtag, empty address, fragment, invented brand);
 - oracle F0.5, and candidates per S1 (mean and p99).
- **The trimmed set is exactly** `candidate_pairs.tsv`: what the matcher scores.

4.4 Pair features (03)

Principles

- No `country` feature.
- Frequency features are rates per split and country, so train and test agree.
- Computed with `rapidfuzz cdist / cpdist` and `numba` over token CSR arrays, stored as `float32`.
- Never a Python loop over pairs.
- Column names follow `<group>__<name>` (C8).

v0 (about 40 features, Friday)

- `name__*`:
 - `ratio`, `token_sort`, `token_set`, `partial` and Jaro-Winkler on `n_core`;
 - token Jaccard, containment both ways, IDF cosine;
 - first- and last-token match, token-count difference;
 - legal-form relation (equal, compatible, conflicting, missing);
 - `n_concat` vs domain/hashtag.
- `extra__*` (the look-alike detector, #9). Tokens in R not in S1, and in S1 not in R, after absorbing typos (edit distance ≤ 2 or JW ≥ 0.9). For each side: count, IDF sum and IDF max, plus a substitution flag.
- `num__*`:
 - the primary-number relation: equal, truncation, prefix, suffix, letter suffix, small nudge ($|d| \leq 10$), or other;
 - absolute and relative difference, digit edit distance, same length;
 - number-set Jaccard and both differences, "every R number compatible", unit equality.
- `addr__*`:
 - street-token Jaccard and IDF overlap, street JW;
 - city equality and JW, state equality;
 - whole-address `token_set` and char TF-IDF cosine;
 - empty, short, landmark, PO box and null flags; component counts.
- `ctx__*`: how many S1 share `n_core` (rate), how many S1 share the (number, street) key (rate), and the name IDF sum.
- `ret__*` and `src__*`: per-view scores and ranks in both directions, view count, and whether R is S2 or S3.

v1 (Saturday)

- **Per-token distractor likelihood:** for each extra or missing token, target-encoded out-of-fold, holdout excluded. Plan A's idea; evidence in #9; gate G3.
- The maximum over name variants: original, OCR, skeleton, dictionary.
- DBA best part, acronym match, Indic dictionary coverage, city alias match.
- Embedding cosines from the SVD views.

4.5 Stage-1 model (04/05)

- **Model.** XGBoost 3.2 `hist`, with `device=cuda` where a GPU exists and `CPU hist` elsewhere. LightGBM is only a blend candidate (gate G10).
- **Training rows.** Candidates of S1 entities in **train folds 5–19** (C2).
 - v0 may train on a sample of those entities. Each sampled entity keeps all its candidates, and the candidates always come from the full record pool.

- **Out-of-fold scores.** Three groups: folds 5–9, 10–14 and 15–19 (`ber.eval.splits.oof_group`). Each group is scored by a model trained on the other two.
 - The holdout and test are scored by the model trained on all of folds 5–19.
 - Early stopping uses a fixed-seed slice.
- **v0 path:** p1, then isotonic calibration cross-fitted on the OOF scores (never on the holdout), then the decision (§4.7).

4.6 Stage-2 collective model (Saturday, gate G4)

- **Scope.** Built on OOF p1 over the **full** candidate graph: every S1 and every record, never a sampled "world". Otherwise the competition features see only a fraction of the real rivals.
- **Features:**
 - *Competition, per record:* the best and second-best p1 among its S1s, the margin to the best rival, this S1's rank for the record, the number of S1 with $p1 > 0.5$, and a mutual-best flag. Plans A and B.
 - *S1 context:* the record's rank among the S1's candidates; counts above 0.5, 0.8 and 0.95; $\Sigma p1$, the expected cluster size; gaps between neighbouring ranks. Plan A.
 - *Source balance:* the number of confident S2 vs S3 matches. Plan A.
- **Owner-removed augmentation (Plan B).** Copies of training record groups with the true owner removed, labelled "no owner". This teaches the model what a missed owner looks like, the most dangerous failure (a confident wrong merge).
- **Cluster support (Sunday, gate G9).** The maximum over the S1's other candidates R' of $p1(S1, R') - \text{sim}(R, R')$. Plan A.
- **Calibration.** Isotonic on OOF p2, cross-fitted. Reliability plots per country and source (Plan B).

4.7 Ownership and decision (06)

- **Ownership.**
 - v0: keep each record only under its highest-p S1 (Plan A). This is the most likely assignment. It is not guaranteed to be expected-F-optimal.
 - Gate G5: a softmax over each record's S1 candidates plus "none" (Plan B) must beat it.
- **Decision.**
 - v0: one global threshold on p, tuned on the holdout.
 - Gate G6: the exact expected-F0.5 per S1 (§2), with one global shrink parameter, must beat the threshold in a paired bootstrap. **Ties go to the threshold** (Plan B's rule).
- **Test-shift check (gate G8).** Compare test with the holdout per country:
 - the histogram of predicted matches per S1;
 - the fraction of records assigned;
 - the distribution of max p per S1;
 - an EM estimate of the positive rate among candidates on test scores.

Shift the threshold or shrink only when these agree, and confirm with at most one leaderboard probe (#13, #14).

- **Outputs** are written through `ber.io` and must get validator PASS.

4.8 Gated extras (Sunday, only with measured gain)

- Learned pre-ranker (Plan A, G11).
- Learned encoder view or feature: multilingual-e5-small, MIT (G10).
- Cross-encoder as a stage-2 feature, run on the uncertain band ($0.1 < p < 0.9$) or the Indic slice: bge-reranker-v2-m3, Apache-2.0, 568M parameters (G10).
- LightGBM blend (G10).
- France pseudo-label adaptation (G7/G8).
- **Rule:** at least +0.002 holdout F0.5 with a 95% CI above 0 (+0.003 for heavy components), no loss in the leave-one-country-out check or the France diagnostics, and the component fits the runtime budget.

5. Validation protocol

1. **Shared holdout** (C2): folds 0–4, 549,699 S1 (US 329,717; India 219,982).
 - Every number is reported there: overall, per country, singleton F0.5, micro precision and recall, and mean predicted per S1 (`ber.eval.report`).
 - Quick iterations may use fold 0 only (about 110k S1, noise about ± 0.002). PR numbers use the full holdout.
2. **Training uses folds 5–19**. Every supervised statistic excludes the holdout: dictionaries, token encodings, city aliases, calibration.
3. **Blocking** always runs over the full train record pool. Stage-2 features always use the full candidate graph.
4. **Gates** use a paired bootstrap over S1 entities (`ber.eval.gates`, 1,000 resamples). Keep a component if $\Delta \geq +0.002$ and the 95% CI is above 0.
 - Every gate result becomes a record in `docs/decisions/` with the baseline tag, the candidate tag, Δ , the CI and the verdict.
5. **Robustness checks**:
 - *Leave one country out*: train on US, score India, and the reverse. Record the drop, and prune features that collapse (G13).
 - *Missed-owner check*: drop 20% of holdout S1 together with their pairs, recompute the stage-2 features, and measure false positives on the records that lost their owner. This is a failure-mode check, **not** a threshold driver (#14).
6. **Test diagnostics** per country (§4.7), plus 100 hand-checked French predictions.
7. **Leaderboard**:
 - a sanity check (a stable gap between validation and leaderboard);
 - one France-vs-empty probe (G7);
 - at most one threshold probe.
 - Never tune several knobs on the public leaderboard.
8. **Ablation table**: remove one component at a time, for the methodology document (Plan B, V10).

6. France (test only, 15% of S1)

- **Evidence**: the same generator as US and India (#15). There are no postcodes (#10). Generic names collide heavily, so rarity and number features carry the decision.
- **Handling**:
 - country-agnostic features only;
 - IDF and TF-IDF fitted on the test France records too;
 - French lexicons:
 - street types (rue/r, avenue/av, boulevard/bd, impasse/imp, chemin/ch, allée, quai, route/rte, faubourg/fg);
 - legal forms (SARL, SAS, SASU, EURL, SA, SCI, SNC, EI);
 - bis/ter, Cedex, arrondissements, and department → region.
- **Checks**:
 - the leave-one-country-out drop, as a fragility estimate;
 - France's max-p distribution and predicted matches per S1 compared with US and India;
 - 100 hand checks;
 - the France-vs-empty leaderboard probe. If the predictions don't beat an empty France, stop and investigate before the final submission.

7. Compute budget and hardware

step	target time	notes
records cache	~1 min	pyarrow

step	target time	notes
normalize 23.7M records	~5 min	24 processes
TF-IDF + SVD, 2 main views × 2 splits (+ small V-name-short)	~20 min	fit on a 2M sample, chunked transform
GPU top-k, 2 main views × 2 splits	~40 min	fp16 tiles, top-k buffers on the GPU
features: v0 about 35M pairs (train sample + holdout + test); v1 full graph about 120M	~30 min / ~70 min	numba CSR, rapidfuzz, float32, chunked
XGBoost stage 1 (3 OOF + full) + stage 2	~20 min	QuantileDMatrix on GPU
decide + write + validate	~3 min	numba DP

- **End to end:** about 2 h, and iterations reuse cached stages.
- **GPU stages** run on Ameya's laptop (RTX 5070 Ti 12 GB, 24 cores, 31 GB). Everything else also runs on the CPU.
- **Teammates:**
 - develop on fold-0 subsets;
 - get shared candidate and feature artifacts from the team drive, with a `MANIFEST.txt` of sha256 hashes (CONTRIBUTING §7).
- **Rules:**
 - process per country;
 - use int32/float32;
 - chunk every stage;
 - guard entry points with `if __name__ == "__main__":` (Windows uses spawn);
 - no faiss-gpu (no Windows wheels), so use torch matmul top-k.

8. Milestones and submissions (IST)

when	milestone	done when
Fri 14:00–15:30	M0: this plan, contracts and roadmap merged; pipeline skeleton merged; everyone set up	<code>pytest</code> green on every machine; <code>python -m ber.pipeline --stage</code> <code>records</code> works
Fri 15:30–18:30	M1: normalize v0; block v0 (keys by ~16:30 so others have real candidates, then V-both + V-addr); features v0 and model/decision plumbing on the key candidates	artifacts for fold 0
Fri 18:30–21:00	M2: full train candidates and the G1 report; features v0 at scale; stage-1 XGBoost; holdout F0.5 with argmax + threshold	a holdout report JSON
Fri 21:00–23:30	M3: test run, outputs, validator	Submission 1 (by 23:30), with a record and a tag
Sat 09:00–13:00	M4: stage 2 + calibration + owner-removed augmentation (G4); DP vs threshold (G6); per-token distractor encoding (G3); V-name-short + K tuning (G1 v1); engine check (G2)	Submissions 2–3
Sat 13:00–18:00	M5: error-analysis gallery, features v1 (variants, Indic dictionary, DBA/acronym; G12), softmax vs argmax (G5), leave-one-country-out and missed-owner checks (G13)	Submission 4
Sat 18:00–23:00	M6: test diagnostics and prior estimate (G8), France checks and probe (G7), threshold choice	Submissions 5–6

when	milestone	done when
Sun 09:00– 15:00	M7: gated extras (G9–G11), final thresholds	Submissions 7–9
Sun 15:00– 17:00	M8: choose the final candidate (holdout, diagnostics, leaderboard consistency); code freeze at 17:00	a decision record
Sun 17:00– 22:00	M9: clean re-run from raw TSV in a fresh environment; README commands and run times; <code>Documentation_template.md</code> ; zip; validator on the zipped outputs	final upload = the chosen model , tag <code>final</code> , zip uploaded; buffer until 23:59

- The detailed tasks and owners are in `docs/ROADMAP.md`.
- Every upload gets a record in `submissions/records/` and a `sub/...` tag. Plan for about 10–11 uploads, and keep one slot per day in reserve.

9. Gates (a hypothesis, the experiment that tests it, the rule)

gate	hypothesis [H]	experiment	keep if	area	when
G1	Blocking reaches $\geq 99.0\%$ (v0) / $\geq 99.5\%$ (v1) pair recall at about 30 candidates per S1	holdout recall per country, source and hard case; recall vs pairs per S1	the target is met; each view adds ≥ 0.1 pt	block	Fri / Sat
G2	SVD-256 dense search $\approx$ sparse exact recall@K	100k-entity sample, the same K	loss ≤ 0.2 pt, else use sparse	block	Fri
G3	Per-token distractor encoding helps	ablation	$\Delta \geq +0.002$, CI > 0	features	Sat
G4	The stage-2 collective model beats stage 1 alone	paired bootstrap	$\Delta \geq +0.002$, CI > 0	model	Sat
G5	Softmax with "none" beats per-record argmax	paired bootstrap	$\Delta \geq +0.002$, else argmax	model	Sat
G6	Expected-F0.5 DP beats a tuned threshold	paired bootstrap	$\Delta > 0$ with CI > 0 ; ties go to the threshold	model	Sat
G7	France predictions beat an empty France	1 leaderboard probe	yes, else investigate	captain + model	Sat
G8	Test needs a threshold/shrink shift	test diagnostics + EM prior	consistent evidence, ≤ 1 probe	model	Sat
G9	Cluster support helps weak direct evidence	ablation	$\Delta \geq +0.002$	model	Sun
G10	Encoder / cross-encoder / LightGBM blend adds value	ablation + runtime	$\Delta \geq +0.003$ and fits the budget	block / model	Sun
G11	A learned pre-ranker is needed	pairs per S1, feature time, recall loss	needed and loses $< 0.1\%$ recall	block	Sat / Sun
G12	Transliteration + dictionary + skeleton lift the Indic slice	Indic-slice F0.5	the slice improves with no global loss	normalize	Sat
G13	Features transfer across countries	leave-one-country-out	the drop is recorded; features that collapse are pruned	model	Sat

10. Team and ownership (proposed; confirmed in the PR review, mirrored in

area	owner	backup
coordination, submissions captain, blocking (GPU runs), pipeline CLI, packaging	Ameya (@AmeyaBorkar)	Sachi
model, calibration, ownership and decision, gates and ablations, error analysis	Sachi (@ssdhoka06)	Ameya
normalization, lexicons (US/India/France, Indic), pair features	Member 3 (@trustdemons05)	Ameya
methodology document	each owner writes their section; the coordinator compiles it	—

These owners match each plan's strengths: Plan A's GPU retrieval and Plan B's validation discipline. The Friday critical path is normalize → features, which is why one person owns both string-heavy stages and the contracts between them.

11. Risks and mitigations

risk	mitigation
Test density differs (+23% records per S1)	test diagnostics + EM prior (G8), competition features, conservative until confirmed
Blocking misses the owner, so the record lands confidently on the wrong S1	the recall gate per hard case, owner-removed augmentation, the missed-owner check
Common-name explosion (#6)	no name-only view over full addresses, rarity features, address tie-breaks in V-both
Look-alikes (#9)	number relations, extra-token features and encodings, competition
France unseen	country-agnostic features, test-fitted statistics, lexicons, leave-one-country-out, the probe
Friday overrun	v0 scope only; every milestone ends in a submission; extras are gated
One GPU machine	CPU fallbacks, cached stages, shared artifacts with manifests
Memory (16–32 GB machines)	per-country processing, compact dtypes, chunking, sampled training entities
Overfitting the public leaderboard	decisions come from the 550k-entity holdout; at most one probe per knob
Compliance	provided data only; MIT/Apache-2.0 models ≤8B; library licenses checked (no GPL, e.g. no unicode)

12. Deliberately not built (and why)

idea	why not
Name-only retrieval over records with full addresses	0.27% of true pairs need it (#7), and names collide 46–54% (#6)
Postcode feature	≤0.5% of addresses have one, France included (#10)
State-partitioned blocking	states are often missing, abbreviated or in regional scripts
country as a feature	breaks on France [F]
Singleton-specific model	singletons are 5.6%; the decision layer handles them
LLM judge, full graph clustering	too slow at this scale; competition + cluster support cover the useful part
"Test ≈ train" assumption (Plan B #20)	disproved (#13)

idea	why not
Cutting the extra-token features as "covered by competition" (Plan B)	disproved: look-alikes often have no rival S1, so the evidence is inside the pair (#9)
Drop-S1 stress test as the threshold driver (Plan A §5.3)	the extra test records aren't missing their owners (#14); the test is kept as a failure-mode check only

13. Compliance and reproducibility

- **Data:** only the provided data. No external lookups, APIs, geocoding or internet data.
 - Lexicons (street types, legal forms, states, regions, departments) are hand-written domain knowledge. They are documented in the methodology.
- **Models:**
 - XGBoost (Apache-2.0), LightGBM (MIT), our own SVD/isotonic models;
 - optionally multilingual-e5-small (MIT) and bge-reranker-v2-m3 (Apache-2.0). Both are ≤8B parameters.
 - Each PR that adds a model states its license.
- **No IDs or row order** are used as signals.
- **Reproducibility:**
 - one CLI (`python -m ber.pipeline --stage all`);
 - fixed seeds;
 - pinned `requirements.txt`;
 - the git commit and command are stored in every artifact and report;
 - the README gives exact commands and run times;
 - git history plus `CHANGELOG.md` plus submission records form the version history.

14. Where each part came from

component	source
Noise catalogue, normalizer design, Indic offset table, learned dictionary, French lexicons	Plan A
Consonant skeleton, OCR variants, domain guard, DBA scored on both sides	Plan B
Multi-view GPU retrieval in both directions, per-view metadata, recall gate and hard-case report	Plan A
Name-only retrieval only for short addresses, the 0.1-pt rule per view, domain/DBA keys, no learned pruner in v0	Plan B (+ data #7)
Pair features, extra-token model with per-token encodings	Plan A (+ data #9)
Name rarity and shared-address counts	Plans A and B
Stage-2 collective with S1 context and source balance	Plan A
Owner-removed augmentation, softmax with "none" as a gated alternative	Plan B
Expected-F0.5 DP	Plan A, under Plan B's "must beat the threshold" rule
Evidence tags, gates, paired bootstrap, reliability plots, ablation table	Plan B
Shared holdout, OOF groups, test diagnostics	repo contracts + Plan A
Shift handling by diagnostics and an EM prior (not a drop-S1 stress test)	data check of 25 Sep (#14)

Version history

version	date (IST)	change
1.0	2026-09-25 14:15	Accepted: Plan A base + Plan B grafts + data checks of 25 Sep